

Graphing Quadratic Functions

Standard Form for a Quadratic Function:

$$y = ax^2 + bx + c; a \neq 0 \quad \text{shape: } \underline{\hspace{2cm}}$$

Since $b \neq 0$, we will need to find the axis of symmetry and vertex algebraically.

Characteristics:

1. Parabola opens up if $\underline{\hspace{1cm}}$; opens down if $\underline{\hspace{1cm}}$
2. Parabola is wider than $y = x^2$ if $\underline{\hspace{1cm}}$; narrower than $y = x^2$ if $\underline{\hspace{1cm}}$
3. x-coordinate of the vertex is $\underline{\hspace{1cm}}$, thus the vertex is $\underline{\hspace{1cm}}$
 - a. Vertex is an absolute minimum if $\underline{\hspace{1cm}}$
 - b. Vertex is an absolute maximum if $\underline{\hspace{1cm}}$
 - c. y-coordinate is the max/min value of the function and the x-coordinate is where the max/min occurs.
4. Axis of symmetry is the $\underline{\hspace{1cm}}$ line $x = \underline{\hspace{1cm}}$
5. Domain is the set of $\underline{\hspace{1cm}}$ values or $\underline{\hspace{1cm}}$ values. For all quadratic functions, the domain is ALWAYS $\underline{\hspace{1cm}}$
6. Range is the set of $\underline{\hspace{1cm}}$ values or $\underline{\hspace{1cm}}$ values. Range is dependent on the y-coordinate of the vertex.

Example 1: Determine whether each quadratic function has an absolute minimum/maximum. Then, find the coordinates of the absolute min/max.

A) $f(x) = 2x^2 + 3x - 1$

B) $f(x) = -\frac{1}{2}x^2 - 3x + 1$

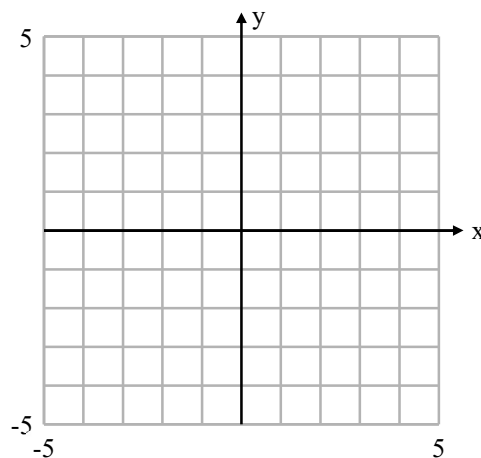
To Graph! (Same steps as graphing absolute value)

1. Plot the vertex
2. Draw the axis of symmetry (DASHED LINE)
3. Plot two points on ONE side of the axis of symmetry
4. Use the axis of symmetry to plot the corresponding two points
5. Draw the parabola

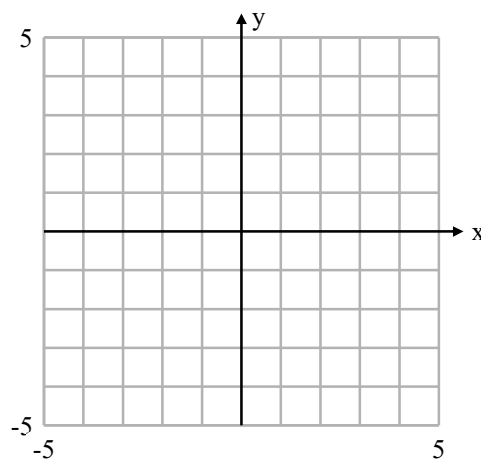
For the examples below, determine if the vertex is an absolute max/min. Find the vertex and axis of symmetry.

State the domain and range. Graph the parabola.

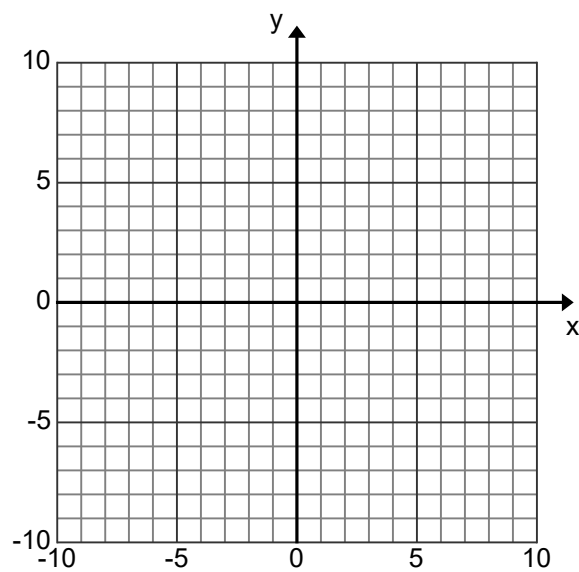
Example 2: Graph $y = \frac{1}{2}x^2 - 3$



Example 3: Graph $y = -x^2 + 2x + 4$



Example 4: Graph $y = 2x^2 + 3x + 1$



Example 5: Graph $y = -\frac{1}{2}x^2 + 4x - 2$

